

GRU Performance Evaluation for Indonesian Inflation

Evaluating the contribution of interest rates and rupiah exchange-rate movements to inflation forecasting



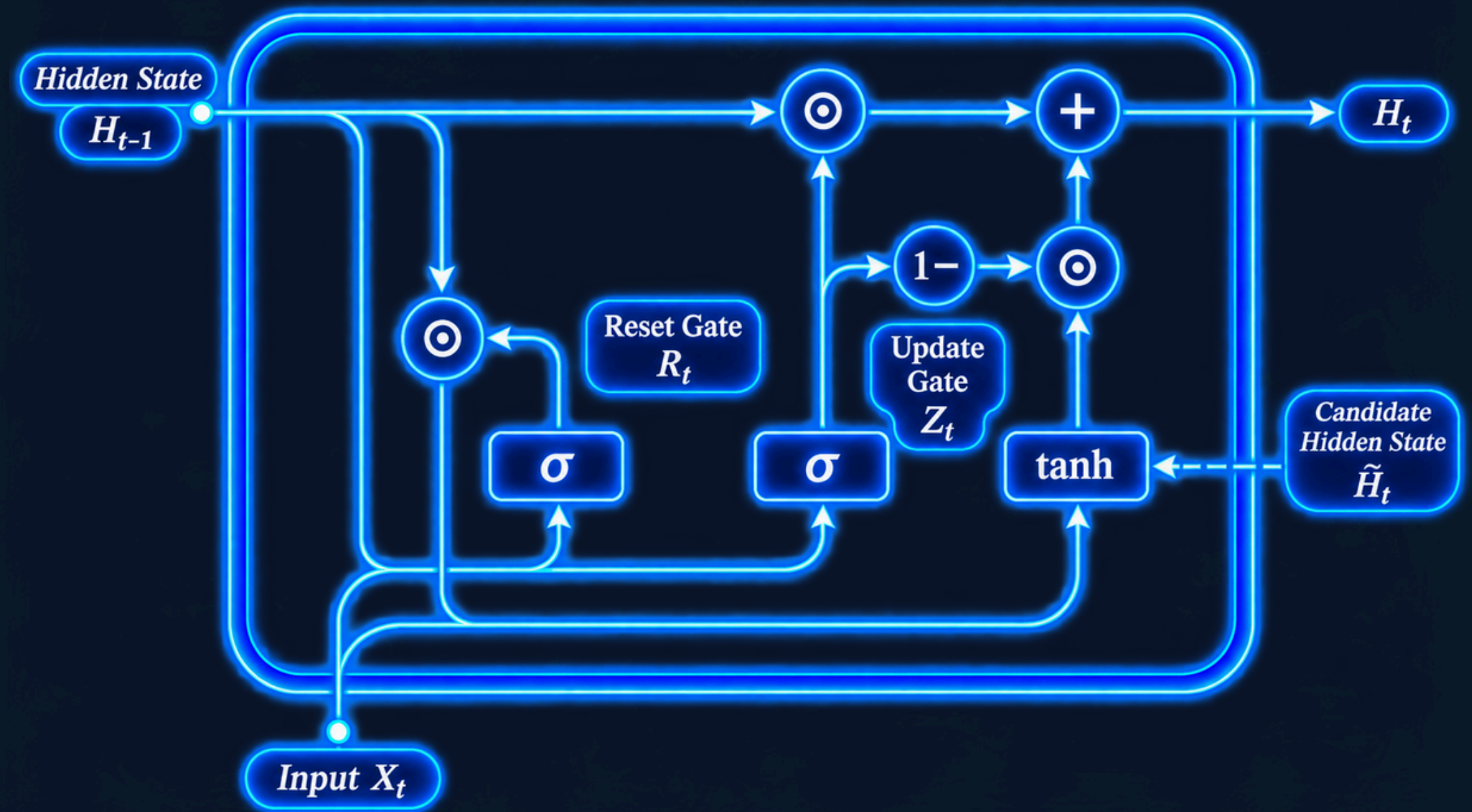
A multivariate time-series forecasting project using Gated Recurrent Unit neural networks.

Time-Series Forecasting

Deep Learning

Macroeconomic Analysis

Gated Recurrent Unit Architecture



monthly data



2009-2025



4 model scenarios



3 evaluation metrics

Why Forecast Inflation?



Problem

Inflation is a key indicator of economic stability. Reliable forecasts support monetary policy formulation and economic decision-making.



Challenge

- Inflation is volatile and shaped by nonlinear relationships among historical patterns, interest rates, and exchange-rate movements.
- Conventional statistical models may be limited in capturing complex nonlinear patterns and long-term temporal dependencies.



Project Objective

To evaluate whether adding interest rates and changes in the Rupiah exchange rate improves GRU-based inflation forecasting compared with using historical inflation alone.



Research Question

Do interest rates and Rupiah exchange-rate changes improve GRU-based inflation forecasting?



Historical Inflation
Interest Rate
Exchange-Rate Change



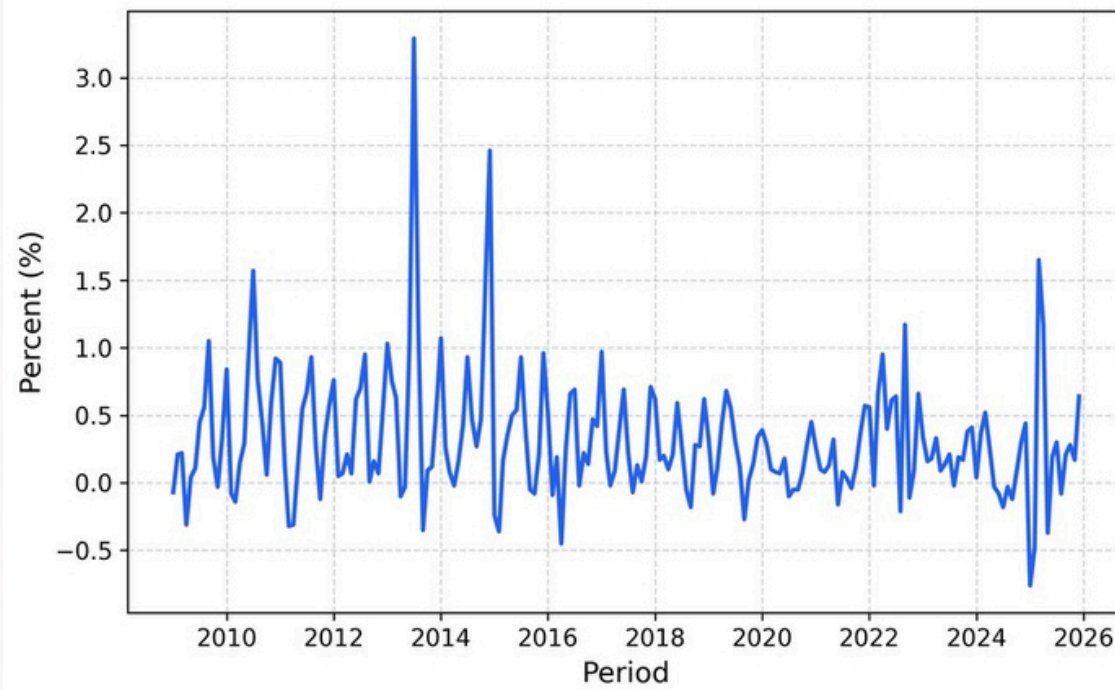
GRU



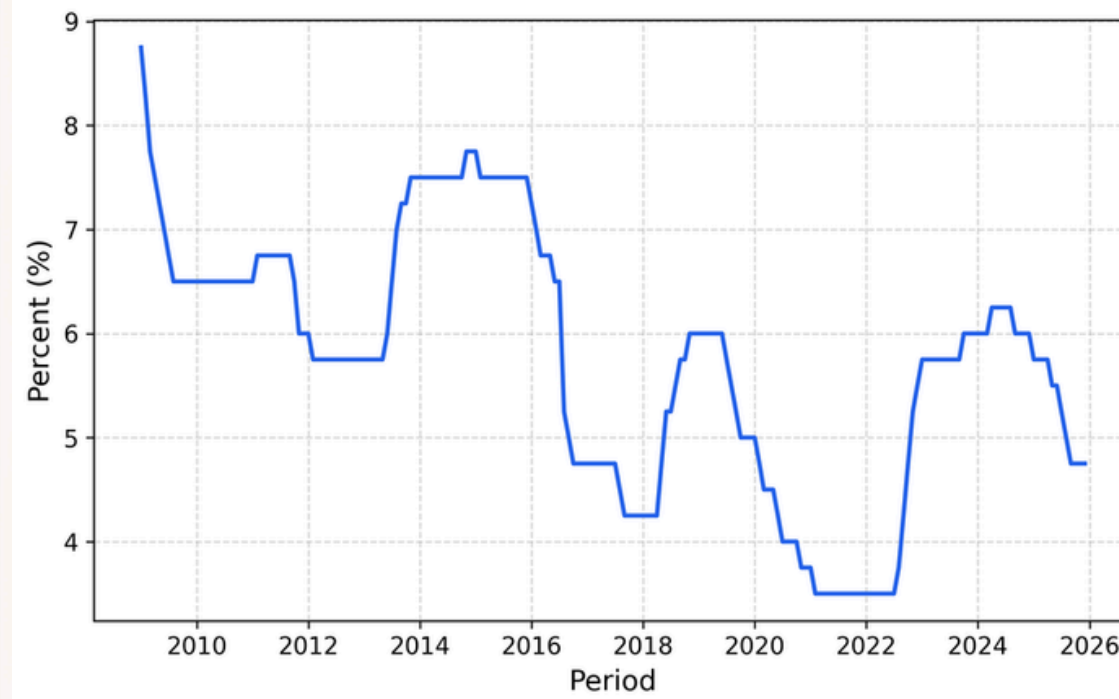
Next-Period
Inflation

Understanding the Macroeconomic Data

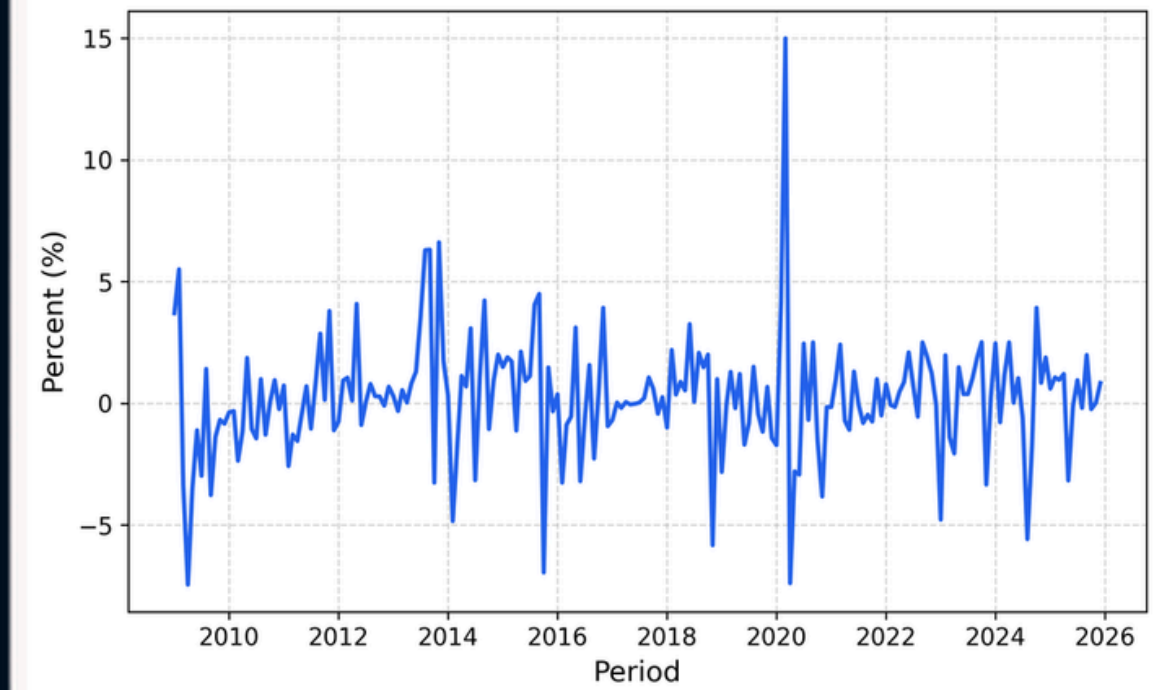
Historical Inflation Data (2009-2025)



Historical Interest Rate Data (2009-2025)



Historical Rupiah Exchange-Rate Change (2009-2025)



Inflation: Mean 0.318% •
Standard Deviation 0.461%



Interest Rate: Mean 5.752%
• Standard Deviation 1.247%



Exchange-Rate Change: Mean
0.239% • Standard Deviation 2.444%



INSIGHT

Exchange-rate changes were the most volatile variable, while interest rates moved more gradually over time.



OBSERVATION

The variables show distinct temporal patterns, fluctuations, and occasional extreme movements.

From Raw Data to Inflation Prediction

01

Data Collection



Monthly inflation, BI interest rate, and USD/IDR exchange-rate data

02

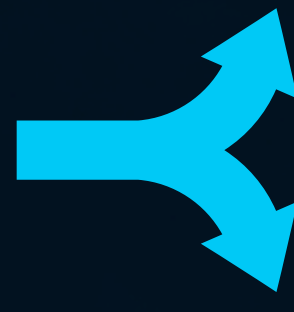
Transformation



Converted the exchange rate into monthly percentage changes

03

Data Splitting



Chronologically split the data into training, validation, and testing sets.

04

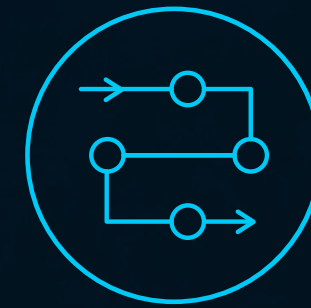
Normalization



Applied Min-Max scaling within a range of -1 to 1

05

Sequence Generation



Created time-series sequences for the GRU input

06

Training and Evaluation



Trained using the Adam optimizer and evaluated with MAE, RMSE, and MAAPE

Monthly Exchange-Rate Change

$$K_t = \left(\frac{N_t - N_{t-1}}{N_{t-1}} \right) \times 100\%$$

K = ExchangeRate Change

N = Exchange Rate
at time t

N_{t-1} = Exchange Rate
at time t - 1

Model Architecture



Model Performance

Evaluating the contribution of interest rates and rupiah exchange rate movements to inflation forecasting

Model	Input	MAE	RMSE	MAAPE
A	Inflation	0.337294	0.485516	0.783335
B	Inflation + Interest Rate	0.327463	0.449823	0.775113
C	Inflation + Exchange-Rate Change	0.330841	0.484603	0.765823
D	All Variables	0.324885	0.457581	0.747259

 Lower metric values indicate better forecasting performance



OBSERVATION

The loss curves shown represent Model D, which uses inflation, interest rate, and exchange-rate change as input variables.

Training and validation losses decreased rapidly during the early epochs and then stabilized.



OBSERVATION

Model D achieved the best overall performance with the lowest MAE and MAAPE, while Model B recorded the lowest RMSE



Key Findings & Conclusion

How additional macroeconomic variables improved GRU-based inflation modeling

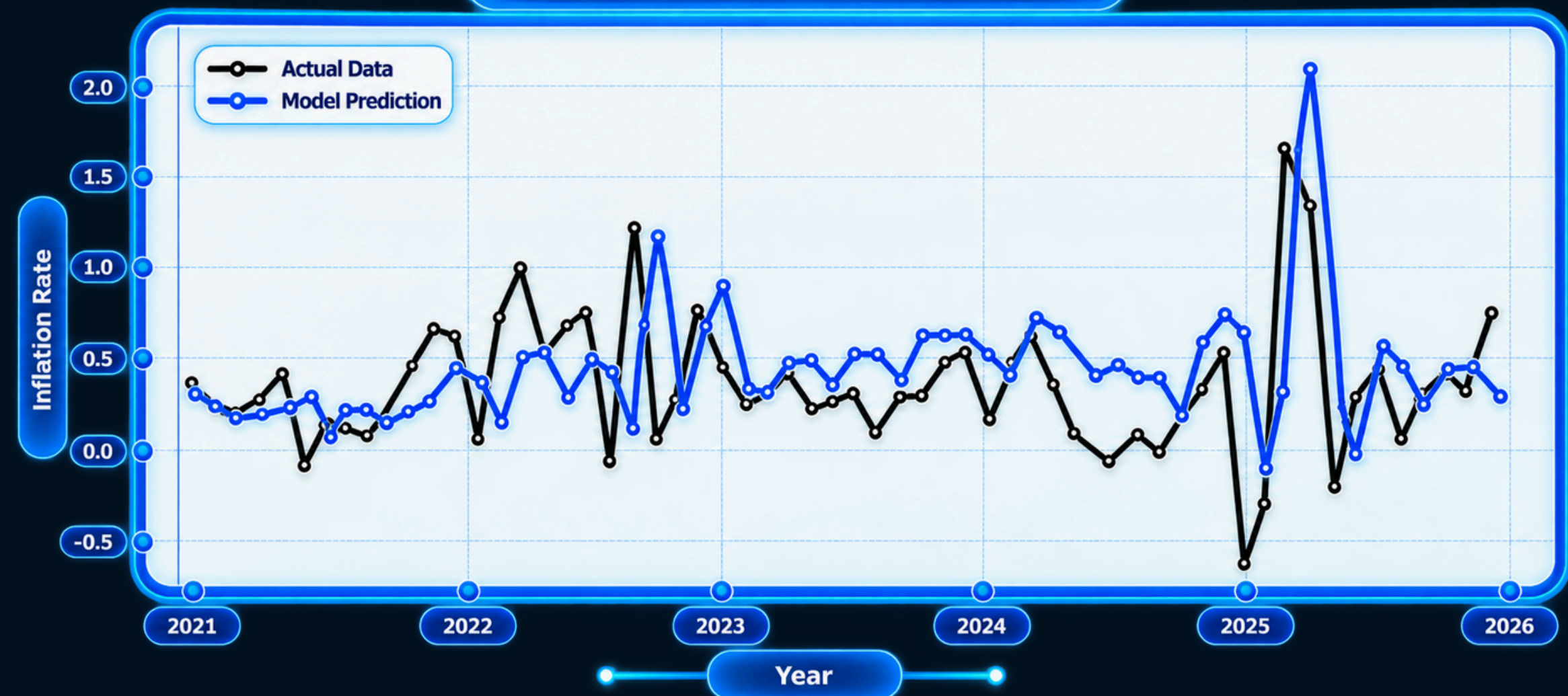
Key Findings

- Additional macroeconomic variables improved overall forecasting performance
- Interest-rate information produced the lowest RMSE when combined with inflation
- Model D achieved the lowest MAE and MAAPE using all three variables

Conclusion

- Demonstrates the potential of deep learning for economic forecasting.
- Highlights the contribution of interest rates and exchange-rate changes to improving prediction accuracy

Actual vs. Predicted Inflation — Model D



Model D captured the overall inflation pattern, although deviations remained during sharp fluctuations